Navier-Stokes 2D Advective + Diffusive parts (Burgers Equations).

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We start from the conservative vectorial form of the 2D Burgers equation (Advective and Diffusive terms):

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) = \nu \nabla^2 \mathbf{u} \tag{1}$$

where $\mathbf{u} = (v_x, v_y)^T$ is the velocity vector and ν is the kinematic viscosity.
Since we are simulating in a 2D domain divided into small control volumes, we want to d
etermine how the velocity field develops inside each cell. We integrate equation (1) over a triangular cell A_i:

$$\iint_{A_i} \frac{\partial \mathbf{u}}{\partial t}\, dA_i + \iint_{A_i} \nabla \cdot (\mathbf{u}\mathbf{u})\, dA_i = \iint_{A_i} \nu \nabla^2 \mathbf{u}\, dA_i \tag{2}$$

Applying the 2D Gauss divergence theorem to the advective and diffusive flux terms, and assuming the velocity vector $\mathbf{u}_i$ represents the average value over the cell area A_i:

$$A_i \frac{\partial \mathbf{u}_i}{\partial t} + \oint_{L_i} (\mathbf{u}\mathbf{u}) \cdot \mathbf{n}\, dl_i = \oint_{L_i} \nu (\nabla \mathbf{u}) \cdot \mathbf{n}\, dl_i \tag{3}$$

Rearranging to isolate the time derivative term:

$$A_i \frac{\partial \mathbf{u}_i}{\partial t} = \oint_{L_i} \left[-(\mathbf{u}\mathbf{u}) + \nu \nabla \mathbf{u}\right] \cdot \mathbf{n}\, dl_i \tag{4}$$

We discretize this spatial integration over the boundary of the unstructured triangular cell. The closed line integral is replaced by a summation over the three edges $j = 1, 2, 3$:

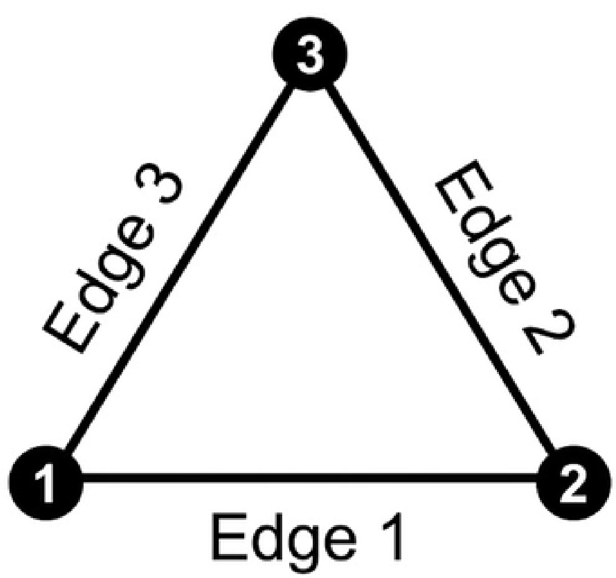

Figure 1: Triangular cell with counter-clockwise numbering of nodes and edges.

$$A_i \frac{\partial \mathbf{u}_i}{\partial t} = \sum_{j=1}^{3} \left[-(\mathbf{u}_{ij}\mathbf{u}_{ij}) \cdot \mathbf{n}_{ij} + \nu (\nabla \mathbf{u})_{ij} \cdot \mathbf{n}_{ij}\right] L_{ij} \tag{5}$$

Approximating the temporal derivative using a first-order explicit Forward Euler scheme (where superindex n denotes the current time step):

$$A_i \frac{\mathbf{u}_i^{n+1} - \mathbf{u}_i^n}{\Delta t} = \sum_{j=1}^{3} \left[-(\mathbf{u}_{ij}^n \mathbf{u}_{ij}^n) \cdot \mathbf{n}_{ij} + \nu (\nabla \mathbf{u})_{ij}^n \cdot \mathbf{n}_{ij} \right] L_{ij} \tag{6}$$

Writing equation (6) in matrix form and solving explicitly for the velocity vector at the next time step $n+1$:

$$\begin{pmatrix} v_{xi} \\ v_{yi} \end{pmatrix}^{n+1} = \begin{pmatrix} v_{xi} \\ v_{yi} \end{pmatrix}^{n} + \frac{\Delta t}{A_i} \sum_{j=1}^{3} \left[- \begin{pmatrix} (v_{xij}^n)^2 & v_{xij}^n v_{yij}^n \\ v_{yij}^n v_{xij}^n & (v_{yij}^n)^2 \end{pmatrix} \begin{pmatrix} n_{xij} \\ n_{yij} \end{pmatrix} + \nu \begin{pmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} \end{pmatrix}_{ij}^{n} \begin{pmatrix} n_{xij} \\ n_{yij} \end{pmatrix} \right] L_{ij} \tag{7}$$

Since equation (7) requires the velocity field evaluated at the edges, but variables are stored at cell centroids, an interpolation scheme is required. Equation (7) also needs the spatial derivatives, which are going to be obtained from a gradient construction scheme.

0.1 Interpolation Schemes On Edges

0.1.1 First Order Upwind

This is used if we have to interpolate velocities on edges from their values in the centroids of the cells that share it. Let i be the target cell ID, k the neighboring cell ID, and j the edge shared between them. The average velocity components at the edge are defined as:

$$\begin{aligned} v_{x,\text{average}}^n &= \frac{v_{xi}^n + v_{xk}^n}{2} \\ v_{y,\text{average}}^n &= \frac{v_{yi}^n + v_{yk}^n}{2} \end{aligned} \tag{8}$$

The position vectors connecting the cell centroids to the edge j are given by:

$$\begin{aligned} \mathbf{n}_{ij} &= (n_{xij}, n_{yij}) \\ \mathbf{n}_{kj} &= (n_{xkj}, n_{ykj}) \end{aligned} \tag{9}$$

Since both vectors are exactly the same but in opposite direction, we can use just one. We have to calculate the inner products of the velocity average and our normal unitary vector, the result will tell us if the fluid is going from i to k or vice versa, the value that will give to the edge is the one from the source cell.

$$\text{If } \mathbf{v}_{\text{average}}^n \cdot \mathbf{n}_{ij} > 0 \quad \Longrightarrow \quad \mathbf{v}_j^n = \mathbf{v}_i^n \tag{10}$$

$$\text{If } \mathbf{v}_{\text{average}}^n \cdot \mathbf{n}_{ij} \leq 0 \quad \Longrightarrow \quad \mathbf{v}_j^n = \mathbf{v}_k^n \tag{11}$$

This interpolation scheme is so stable, easy and fast to add and use but adds a lot of numerical diffusion, which can modify our results very much.

0.1.2 Gradient-Based Skewness and Non-Orthogonality Corrected Interpolation Scheme

This scheme is useful for interpolating spatial variables on mesh edges. First, it is important to consider that the distances from the edge midpoint to the centroids of the adjacent cells are generally not equal. Moreover, the vectors connecting the centroids to the edge midpoint are often not orthogonal to the edge itself. Performing a simple arithmetic average under these conditions introduces significant numerical error.

This technique mitigates these errors by averaging the first-order Taylor series expansions evaluated from each cell centroid to the edge midpoint. Defining $\mathbf{r}_{ij}$ and $\mathbf{r}_{kj}$ as the displacement vectors, and $\nabla \mathbf{v}_i$ and $\nabla \mathbf{v}_k$ as the velocity gradients at the respective cell centroids, the extrapolated values at the edge from each cell are:

$$\mathbf{v}_{ji} = \mathbf{v}_i + \nabla \mathbf{v}_i \cdot \mathbf{r}_{ij} \tag{12}$$

$$\mathbf{v}_{jk} = \mathbf{v}_k + \nabla \mathbf{v}_k \cdot \mathbf{r}_{kj} \tag{13}$$

The final value assigned to the edge is the average of these two extrapolated states:

$$\mathbf{v}_j = \frac{\mathbf{v}_{jk} + \mathbf{v}_{ji}}{2} \tag{14}$$

0.1.3 Linear Geometric-Weights Based Interpolation Scheme

We are going to ignore the fact that the line that unites the centroids does not cross the edge midpoint, but considering that the distances from the midpoint to the centroids are not equal, we are going to obtain the spatial derivatives in the edge midpoint using a ponderate average. Let d_{ij} and d_{kj} be those distances, so d_{ik} is the distance between centroids and can be computed in this way:

$$d_{ik} = d_{ij} + d_{kj} \tag{15}$$

The geometric weights can be obtained as following (considering that the weight is greater as the distance to the midpoint is closer):

$$w_{ij} = \frac{d_{kj}}{d_{ik}} \tag{16}$$

$$w_{kj} = \frac{d_{ij}}{d_{ij}} \tag{17}$$

And we can obtain the spatial derivatives with them:

$$\left(\frac{\partial v_x}{\partial x}\right)_j = w_{ij}\left(\frac{\partial v_x}{\partial x}\right)_i + w_{kj}\left(\frac{\partial v_x}{\partial x}\right)_k \tag{18}$$

$$\left(\frac{\partial v_x}{\partial y}\right)_j = w_{ij}\left(\frac{\partial v_x}{\partial y}\right)_i + w_{kj}\left(\frac{\partial v_x}{\partial y}\right)_k \tag{19}$$

$$\left(\frac{\partial v_y}{\partial x}\right)_j = w_{ij}\left(\frac{\partial v_y}{\partial x}\right)_i + w_{kj}\left(\frac{\partial v_y}{\partial x}\right)_k \tag{20}$$

$$\left(\frac{\partial v_y}{\partial y}\right)_j = w_{ij}\left(\frac{\partial v_y}{\partial y}\right)_i + w_{kj}\left(\frac{\partial v_y}{\partial y}\right)_k \tag{21}$$

0.2 Gradient Construction Schemes

0.2.1 Least Squares

Using the values of velocity in centroids of neighbor cells, we adjust a plane using least squares method. Lets write a Taylor first order expansion:

$$\mathbf{v}_i = \mathbf{v}_j + \nabla \mathbf{v}_i \cdot \mathbf{r}_{ij} \tag{22}$$

Remebering that every domain cell has three neighbors (we are using the ghost cells scheme), lets expand this equation onto 6:

$$\Delta v_{x,i1} = \left(\frac{\partial v_x}{\partial x}\right)_i \Delta x_{i1} + \left(\frac{\partial v_x}{\partial y}\right)_i \Delta y_{i1} \tag{23}$$

$$\Delta v_{x,i2} = \left(\frac{\partial v_x}{\partial x}\right)_i \Delta x_{i2} + \left(\frac{\partial v_x}{\partial y}\right)_i \Delta y_{i2} \tag{24}$$

$$\Delta v_{x,i3} = \left(\frac{\partial v_x}{\partial x}\right)_i \Delta x_{i3} + \left(\frac{\partial v_x}{\partial y}\right)_i \Delta y_{i3} \tag{25}$$

$$\Delta v_{y,i1} = \left(\frac{\partial v_y}{\partial x}\right)_i \Delta x_{i1} + \left(\frac{\partial v_y}{\partial y}\right)_i \Delta y_{i1} \tag{26}$$

$$\Delta v_{y,i2} = \left(\frac{\partial v_y}{\partial x}\right)_i \Delta x_{i2} + \left(\frac{\partial v_y}{\partial y}\right)_i \Delta y_{i2} \tag{27}$$

$$\Delta v_{y,i3} = \left(\frac{\partial v_y}{\partial x}\right)_i \Delta x_{i3} + \left(\frac{\partial v_y}{\partial y}\right)_i \Delta y_{i3} \tag{28}$$

We have to find the spatial derivatives, first of all, lets derive least squares scheme. If we have a set of points in R^3 defined by their coordinates:

$$(x_1, y_1, z_1)$$
$$(x_2, y_2, z_2)$$
$$\vdots$$
$$(x_i, y_i, z_i)$$

We could adjust a plane which equation has this form:

$$z = ax + by \tag{29}$$

If we suppose that a and b makes that equation (19) fits well our points, we can calculate the error of our equation on each point with the difference between the original z_i and the one obtained with the equation:

$$E = z_i - ax_i - by_i \tag{30}$$

We are interested in the distance and not in the sign, so we are going to square it, we do not use the absolute value because it is difficult to derivate it (thing that we are going to do in the next steps). Then we are going to sum every squared error:

$$E_{\text{sum}}^2 = \sum (z_i - ax_i - by_i)^2 \tag{31}$$

If we derivate this cuantity over a and b and put into a equations system (where a and b minimizes the sum):

$$\begin{pmatrix} \sum x_i^2 & \sum x_i y_i \\ \sum x_i y_i & \sum y_i^2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \sum x_i z_i \\ \sum y_i z_i \end{pmatrix} \tag{32}$$

We have the geometric and velocity data (equations (13) to (18)), so lets substitute it in the equations system (22). This will give us two systems:

$$\begin{pmatrix} \sum \Delta x_{ij}^2 & \sum \Delta x_{ij} \Delta y_{ij} \\ \sum \Delta x_{ij} \Delta y_{ij} & \sum \Delta y_{ij}^2 \end{pmatrix} \begin{pmatrix} \left(\frac{\partial v_x}{\partial x}\right)_i \\ \left(\frac{\partial v_x}{\partial y}\right)_i \end{pmatrix} = \begin{pmatrix} \sum \Delta x_{ij} \Delta v_{x,ij} \\ \sum \Delta y_{ij} \Delta v_{x,ij} \end{pmatrix} \tag{33}$$

$$\begin{pmatrix} \sum \Delta x_{ij}^2 & \sum \Delta x_{ij} \Delta y_{ij} \\ \sum \Delta x_{ij} \Delta y_{ij} & \sum \Delta y_{ij}^2 \end{pmatrix} \begin{pmatrix} \left(\frac{\partial v_y}{\partial x}\right)_i \\ \left(\frac{\partial v_y}{\partial y}\right)_i \end{pmatrix} = \begin{pmatrix} \sum \Delta x_{ij} \Delta v_{y,ij} \\ \sum \Delta y_{ij} \Delta v_{y,ij} \end{pmatrix} \tag{34}$$

These are going to be solved for each cell:

$$\left(\frac{\partial v_x}{\partial x}\right)_i = \frac{\left(\sum_j \Delta y_{ij}^2\right)\left(\sum_j \Delta x_{ij} \Delta v_{x,ij}\right) - \left(\sum_j \Delta x_{ij} \Delta y_{ij}\right)\left(\sum_j \Delta y_{ij} \Delta v_{x,ij}\right)}{\left(\sum_j \Delta x_{ij}^2\right)\left(\sum_j \Delta y_{ij}^2\right) - \left(\sum_j \Delta x_{ij} \Delta y_{ij}\right)^2} \tag{35}$$

$$\left(\frac{\partial v_x}{\partial y}\right)_i = \frac{\left(\sum_j \Delta x_{ij}^2\right)\left(\sum_j \Delta y_{ij} \Delta v_{x,ij}\right) - \left(\sum_j \Delta x_{ij} \Delta y_{ij}\right)\left(\sum_j \Delta x_{ij} \Delta v_{x,ij}\right)}{\left(\sum_j \Delta x_{ij}^2\right)\left(\sum_j \Delta y_{ij}^2\right) - \left(\sum_j \Delta x_{ij} \Delta y_{ij}\right)^2} \tag{36}$$

$$\left(\frac{\partial v_y}{\partial x}\right)_i = \frac{\left(\sum_j \Delta y_{ij}^2\right)\left(\sum_j \Delta x_{ij} \Delta v_{y,ij}\right) - \left(\sum_j \Delta x_{ij} \Delta y_{ij}\right)\left(\sum_j \Delta y_{ij} \Delta v_{y,ij}\right)}{\left(\sum_j \Delta x_{ij}^2\right)\left(\sum_j \Delta y_{ij}^2\right) - \left(\sum_j \Delta x_{ij} \Delta y_{ij}\right)^2} \tag{37}$$

$$\left(\frac{\partial v_y}{\partial y}\right)_i = \frac{\left(\sum_j \Delta x_{ij}^2\right)\left(\sum_j \Delta y_{ij} \Delta v_{y,ij}\right) - \left(\sum_j \Delta x_{ij} \Delta y_{ij}\right)\left(\sum_j \Delta x_{ij} \Delta v_{y,ij}\right)}{\left(\sum_j \Delta x_{ij}^2\right)\left(\sum_j \Delta y_{ij}^2\right) - \left(\sum_j \Delta x_{ij} \Delta y_{ij}\right)^2} \tag{38}$$